

大学物理作业 4.

胡教材: P120 5-1, 5-5, 5-6, 5-7, 5-9, 5-10

吴教材: P155 4-1, 4-6, 4-10.

系列化 P54 4-2, 4-8, 4-13

胡教材 P120 5-1: (1) $I = \int_0^1 F dt = \int_0^1 F_0 \cos(2\pi t) dt = \frac{1}{2\pi} F_0 \sin(2\pi t) \Big|_0^1 = 0$

(2) $v(t) = \int_0^t \frac{F(t)}{m} dt = \frac{F_0}{m} \int_0^t \cos(2\pi t) dt = \frac{F_0}{2\pi m} \sin(2\pi t)$

$\therefore W = \int_0^1 F(t)v(t) dt = \int_0^1 \frac{F_0^2}{4\pi^2 m} \sin(4\pi t) dt = 0.$

(3) $x(t) = \int_0^t v(t) dt = \frac{F_0}{2\pi m} \int_0^t \sin(2\pi t) dt = -\frac{F_0}{4\pi^2 m} \cos(2\pi t) \Big|_0^t = \frac{F_0}{4\pi^2 m} (1 - \cos(2\pi t))$

质点做简谐振动, $x(t) = \frac{F_0}{4\pi^2 m} (1 - \cos(2\pi t))$

5-5: 以向右为正, 由于碰撞时间极短, 我们认为碰撞前后动量守恒.

由题: 对物体 B, 设碰后速度为 v_B $\therefore -\mu mgl = 0 - \frac{1}{2}mv_B^2 \therefore v_B = \sqrt{2\mu gl}$. (动能定理)

由动量守恒定律: $mv_0 + 0 = mv_A + mv_B \therefore v_A = v_0 - v_B = v_0 - \sqrt{2\mu gl}$.

$\therefore E_{\text{损}} = -\Delta E = \frac{1}{2}mv_0^2 - \frac{1}{2}mv_B^2 - \frac{1}{2}mv_A^2 = \frac{1}{2}mv_0^2 - \mu mgl - \frac{1}{2}m(v_0 - \sqrt{2\mu gl})^2 = m\mu\sqrt{2\mu gl} - 2\mu mgl$.

5-6. 1° 相对速度. 以向右为正, 设分离速度为 v , 由动量定理: $MAv_0 + 0 = (MA + MB)v \therefore v = \frac{MA}{MA + MB} v_0$

2° 由动能定理: $W_f = \Delta E_k = \frac{1}{2}(MA + MB)v^2 - \frac{1}{2}MAv_0^2 = \frac{1}{2}(MA + MB) \frac{MA^2}{(MA + MB)^2} v_0^2 - \frac{1}{2}MAv_0^2 = -\frac{1}{2} \frac{MA MB}{MA + MB} v_0^2$

5-7. 由于碰撞发生瞬间, 小对仍在对地静止, $\therefore$ 由动量定理与机械能守恒, 不难推知 $v = \bar{v}$.

为了方便问题的求解: 我们采用质心系. 由于系统在水平方向上不受力, 我们不难得知本参考系为惯性系 $v_c = \frac{M}{M+m} v$.

记 M 在此后的位移为 $x_1(t)$. 同理有 $x_2(t)$.

不难得出系表达式:
$$\begin{cases} Mx_1(t) + mx_2(t) = 0, & x_2(t) - x_1(t) = L \sin \theta & (\text{几何关系}) \\ \text{由牛顿运动定律} & \begin{cases} m\ddot{x}_1(t) = -T \sin \theta, \\ m\ddot{y}_1(t) = T \cos \theta - mg \\ \frac{1}{2}m\dot{x}_1^2(t) + \frac{1}{2}m\dot{x}_2^2(t) + mgy_1(t) = \text{Const} \end{cases} \end{cases}$$
 可以解得: $2(L + m \sin^2 \theta) \ddot{\theta} + L \sin \theta \cos \theta \cdot \dot{\theta} + (M + m)g \sin \theta = 0$

至于最大角度, 此时系内所有动能都会转化为 $E_p \therefore \frac{1}{2} \frac{Mm}{M+m} v^2 = mgL(1 - \cos \theta) \therefore \theta_m = \arccos(1 - \frac{Mv^2}{2(M+m)gL})$.

5-9. 我们仅考虑从最高点到落地的过程: $t = \frac{v_0 \sin \theta}{g}$ $v_{x0} = v_0 \cos \theta$

在地中: $\begin{cases} v_m = v_m - u \\ (M+m)v_0 \cos \theta = Mv_m + m(v_m - u) \end{cases} \therefore v_m = v_0 \cos \theta + \frac{m}{M+m} u \therefore \Delta x = \Delta v \cdot t = (\frac{m}{M+m} u) \cdot (\frac{v_0 \sin \theta}{g}) = \frac{mu v_0 \sin \theta}{(M+m)g}$

$\therefore \Delta x = \frac{mu v_0 \sin \theta}{(M+m)g}$

5-10. (1) 由动量守恒与机械能守恒: $mv_1 + MV_1 = 0 \quad \frac{1}{2}kx^2 = \frac{1}{2}mv_1^2 + \frac{1}{2}MV_1^2 \Rightarrow \begin{cases} v_1 = \frac{\sqrt{110}}{11} \\ V_1 = -\frac{\sqrt{110}}{110} \end{cases}$

(2) 问题 (2) 与问题 (1) 方程相同, 但方向发生反向. $\begin{cases} v_2 = -\frac{\sqrt{110}}{11} \\ V_2 = \frac{\sqrt{110}}{110} \end{cases}$

吴教材 P155 4-1. (1) $\vec{v} = \frac{d}{dt} \vec{r} = -wa \sin wt \hat{i} + wb \cos wt \hat{j}$

(2) $I = 0$.

4-6. (1) 质心加速度的最大值在初始取得: $a_c = \frac{F_B}{M} = \frac{1}{m_1 + m_2} k \Delta l$.

(2) 当弹簧第一次恢复原长时质心达到并保持最大速度. $\frac{1}{2}mv^2 = \frac{1}{2}k\Delta l^2 \therefore v_B = \Delta l \sqrt{\frac{k}{m}} \therefore v_{cm} = \frac{P_B}{m_1 + m_2} = \frac{m_2}{m_1 + m_2} \cdot \Delta l \sqrt{\frac{k}{m}}$.

4-10. (1) 等价于在水平方向上发生了一次完全非弹性碰撞: $v' = \frac{m_1 v_0}{m_1 + m_2}$

$\therefore m_2 gh = \frac{1}{2}m_1 v_0^2 - \frac{1}{2}(m_1 + m_2) v'^2 \Rightarrow h = \frac{1}{2} \frac{m_1}{m_1 + m_2} \cdot \frac{v_0^2}{g}$

(2) 等价于发生一次弹性正碰 (高中背过公式, 懒得再推一次)

$v_1 = \frac{m_1 - m_2}{m_1 + m_2} v_0, v_2 = \frac{2m_1}{m_1 + m_2} v_0$.

系列化 P54. 4-2: (1) $I = 0$, 即冲量为 0

(2) $T = \frac{2\pi}{\omega} \quad I_T = mgT = mg \cdot \frac{2\pi}{\omega}$, 方向与重力方向相反.

4-8 $\begin{cases} v_{Ax} = 7.5 \text{ m/s} \\ v_{Ay} = -3 \text{ m/s} \end{cases} \quad \begin{cases} v_{Bx} = 2 \text{ m/s} \\ v_{By} = 4.5 \text{ m/s} \end{cases} \quad \text{由动量守恒: } \begin{cases} m_A v_{Ax} + m_B v_{Bx} + m_C v_{Cx} = (m_A + m_B + m_C) v. \\ m_A v_{Ay} + m_B v_{By} + m_C v_{Cy} = 0. \end{cases}$

解得 $\begin{cases} v_{Cx} = 6.5 \text{ m/s} \\ v_{Cy} = -4.5 \text{ m/s} \end{cases} \therefore C(13, -9)$

4-13. $kx_0 = m_1 g \Rightarrow k = \frac{m_1 g}{x_0}$

$m_2 gh = \frac{1}{2}m_2 v_0^2 \Rightarrow v_0 = \sqrt{2gh}$.

$m_2 v_0 = (m_1 + m_2) V \Rightarrow V = \frac{m_2}{m_1 + m_2} \sqrt{2gh}$

$\frac{1}{2}(m_1 + m_2) V^2 + (m_1 + m_2) g x = \frac{1}{2}k(x_0 + x)^2 - \frac{1}{2}kx_0^2 \Rightarrow \begin{cases} x_1 = 0.3 \\ x_2 = -0.1 \text{ (舍)} \end{cases} \therefore \text{最大距离为 } 0.3 \text{ m.}$

补 1: 由题: 系统在水平方向上动量守恒: $\therefore m_A v_{Ax} + m_B v_{Bx} + m_C v_{Cx} = 0$ 对任意时刻都成立

$\therefore m_A x_A + m_B x_B + m_C x_C = a \quad x_A = 5 - 10\sqrt{3} \quad x_B = 5 - 10, x_C = 5 \quad \therefore 5 = \frac{40\sqrt{3} + 10}{21}$

补 2: (1) m_2 先到

(2) 由动量守恒: 设船速为 v , 以 m_1 的碰前速度方向为正方向: $mv + m_2(v + u) + m_1(v - u) = 0 \therefore v = \frac{(m_1 - m_2)u}{m + m_1 + m_2}$

$\therefore t = \frac{l/2}{\frac{m + 2m_1}{m + m_1 + m_2} u}$